

C Programming Assignment 1

Q1. Display Student Information

Implement a C program to store and display the following information using appropriate primitive data types:

- Student ID
- Age
- Percentage
- Grade

Display all the information clearly using printf().

Q2. Accept and Display User Details

Implement a C program to accept the following details from the user using scanf():

- Age
- Height (in meters)
- Grade (single character)

Display all the entered values clearly using printf().

Q3. Simple Bill Calculation

Implement a C program to calculate the total cost of an item.

The program should:

- Accept the price of one item.
- Accept the quantity purchased.
- Calculate and display the total bill.

Formula:

Total Bill = Price × Quantity

Q4. Calculate Average of Three Numbers

Implement a C program to accept three integer values and calculate their average.

Ensure that the average is displayed as a decimal value, even when all input values are integers.

Example:

Input: 10 15 20

Output: Average = 15.00

Q5. Swap Two Numbers

Implement a C program to accept two integer values and swap their values using a third variable.

Display the values:

- Before swapping
- After swapping

Example:

Before swapping: 10 20

After swapping: 20 10

Q6. Calculate Quotient and Remainder

Implement a C program to accept two integers and calculate:

- Quotient
- Remainder

Use appropriate arithmetic operators and display both results.

Example:

Input: 17 5

Quotient = 3

Remainder = 2

Q7. Calculate Total Marks and Percentage

Implement a C program to accept marks obtained in five subjects, where each subject is out of 100.

Calculate and display:

- Total Marks
- Percentage

Ensure that the percentage is displayed correctly as a decimal value.

Q8. Employee Salary Calculation

An employee's monthly salary consists of:

- Basic Salary
- Allowance
- Bonus

Implement a C program to accept these values and calculate the final salary.

Formula:

Final Salary = Basic Salary + Allowance + Bonus

Display the final salary up to **two decimal places**.

Q9. Temperature Conversion

Implement a C program to accept the temperature in Celsius and convert it into Fahrenheit.

Formula:

Fahrenheit = (Celsius $\times$ 9 / 5) + 32

Ensure that the output is displayed as a decimal value.

Example:

Input: 25

Output: 77.00 °F

Q10. Industry-Style Invoice Calculator

A company wants to generate a simple invoice for a customer.

Implement a C program to accept:

- Product ID
- Product Price
- Quantity
- Discount Percentage

Calculate the following:

Subtotal = Price $\times$ Quantity

Discount Amount = Subtotal $\times$ Discount Percentage / 100

Final Amount = Subtotal – Discount Amount

Display the following details in a properly formatted manner:

Product ID

Subtotal

Discount Amount

Final Payable Amount